

Nikolaos Ziakos

PhD Candidate in Aerospace Engineering |
Composite Airframe Optimization | Surrogate-
Based Design

Madrid, Spain
nickziakos@gmail.com
+30 695 564 4828
[LinkedIn](#)

RESEARCH PROFILE

PhD candidate at Universidad Carlos III de Madrid (UC3M) and doctoral researcher at the Aeroelastic and Structural Design (ASD) Lab. Research focuses on efficient design optimization of composite airframe structures by combining automated parametric finite-element model generation, structural sizing optimization, surrogate modeling, and constrained Bayesian optimization. Current work addresses high-dimensional mixed-variable optimization of airframes under structural constraints including strain/failure indices, stress limits, and buckling eigenvalues, using scalable Gaussian Process and Bayesian Optimization methods. Potential future extensions involve stacking-sequence optimization, improved composite material modeling, physics-informed/multi-fidelity learning, and inclusion of aeroelastic loads.

RESEARCH INTERESTS

Surrogate-assisted optimization

Gaussian processes, Bayesian optimization, trust-region BO, SCBO/TuRBO-type methods, sparse variational GPs

Aircraft structural modeling

Parametric GFEM generation, wingbox/fuselage layouts, automated FE-based analysis workflows

Composite structural optimization

Sizing-layout optimization, laminate parametrization, failure/strain constraints, buckling constraints

Potential future directions

Gradient-enhanced methods, multi-fidelity/physics-informed ML, stacking sequence, aeroelastic constraints

EDUCATION

PhD Candidate in Aerospace Engineering, Universidad Carlos III de Madrid | Sep 2024 - present

Research title: Efficient design optimization of composite airframes enhanced by ML techniques

- Developing an automated framework for layout and sizing optimization of composite airframe structures using detailed GFEM models and FE-based structural analyses.
- Nested surrogate-assisted composite optimization of wingbox with gradient-based inner loop
- High-dimensional sparse surrogate modeling and constrained Bayesian optimization for layout-sizing optimization of composite wingbox.
- Doctoral researcher in the ASD Lab, funded by the Spanish national project TIFON for research on high-fidelity topology and sizing optimization of aircraft substructures integrating AI/ML methods.

Integrated MEng / Diploma in Mechanical Engineering and Aeronautics, University of Patras | Sep 2019 - Jul 2024

GPA: 9.08/10; ranked 1st in the 2024 graduating class

Diploma thesis: Aeroelastic tailoring of high aspect-ratio wings via tow-steering.

- Developed an FE/DLM model of the NASA uCRM-13.5 wing in MSC Patran/Nastran and performed aeroelastic and structural analyses using a MATLAB-based optimization framework.
- Optimized tow-steered composite laminates under multidisciplinary constraints including structural mass, failure, buckling, and flutter-related requirements, comparing variable-stiffness and conventional composite designs.

SELECTED ONGOING METHODOLOGICAL WORK

Nested surrogate-based optimization: Using surrogate models to guide structural layout exploration while gradient-based sizing optimization evaluates the mass and constraint performance of each candidate layout.

High-dimensional Scalable Constrained BO / trust-region BO: Developing feasibility-aware trust-region strategies using scalable sparse surrogate models to solve the constrained coupled layout-sizing optimization problem.

Planned extensions: Inclusion of stacking-sequence variables, gradient-enhanced or multi-fidelity optimization, more realistic composite constraints, and improved aerodynamic/aeroelastic load modeling.

PUBLICATIONS AND RESEARCH OUTPUTS

Journal article

Ziakos, N.; Kilimtzhidis, S.; Kostopoulos, V. "Multidisciplinary aeroelastic optimization of high aspect-ratio composite aircraft wings via tow-steering." CEAS Aeronautical Journal, 2025. DOI: 10.1007/s13272-025-00851-0.

- Developed and optimized tow-steered composite wing laminates for a high-aspect-ratio aircraft wing under aeroelastic and structural constraints.
- Combined FE structural modeling, DLM-based aeroelastic analysis, and MATLAB optimization to assess mass reduction and constraint satisfaction compared with conventional laminates.

Conference papers / proceedings

Ziakos, N.; Cini, A. "Parametric Framework for Automated Structural Layout Modeling and Optimization of Composite Airframe." AIAA SciTech 2026 Forum, 2026. DOI: 10.2514/6.2026-0586. Presented.

- Presents an automated parametric framework for the nested layout-sizing optimization of composite airframe, with gradient-based sizing and surrogate-assisted layout exploration, supporting early-stage design acceleration.

Ziakos, N.; Cini, A. "Automated Parametric Finite Element Model Generation and Optimization of a Composite Aircraft Wing." Engineering Proceedings (MDPI), EASN Conference 2025, accepted for publication. Presented.

- Describes automated parametric FE model generation and optimization of composite aircraft wings and fuselage segments, forming the basis for the later nested surrogate-based optimization workflow.

TECHNICAL SKILLS

Optimization and ML:	Surrogate modeling; Gaussian processes; Bayesian optimization; constrained optimization; gradient-based optimization; trust-region BO; SCBO/TurBO-type methods; Design of Experiments; neural networks and physics-informed ML concepts.
Structural analysis and composites:	Composite materials; laminate mechanics; aeroelasticity; buckling and failure constraints; structural sizing/topology optimization; aircraft wingbox/fuselage structural modeling.
FEA / CAD / engineering software:	Altair HyperMesh/OptiStruct; MSC Nastran/Patran; Ansys APDL; CATIA V5; AutoCAD.
Programming and scripting:	Python; MATLAB; HyperMesh TCL; C; LaTeX; Microsoft Office.

HONORS AND AWARDS

- Ranked 1st among all graduates of the Department of Mechanical Engineering and Aeronautics, University of Patras, 2024.
- IKY State Scholarships Foundation / Ministry of Education Excellence Award (Aristeio), Greece, 2025.
- University of Patras / ELKE Excellence Commendation (Epainos) for outstanding academic performance, 2025.

ADDITIONAL EXPERIENCE

Cooling Team Member, UoP Racing Team, University of Patras | Oct 2021 - Jan 2022

Designed cooling-system components in CATIA V5 and collaborated with the Formula Student team on design improvements.

Student organizations | 2021 - 2024

Active member of EESTEC LC Patras, EUROAVIA Patras, and BEST Patras; former IT member and rocket-team member in student engineering organizations.

LANGUAGES

Greek: native | English: C2 | Spanish: B2 | German: B1